

Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

CODE:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

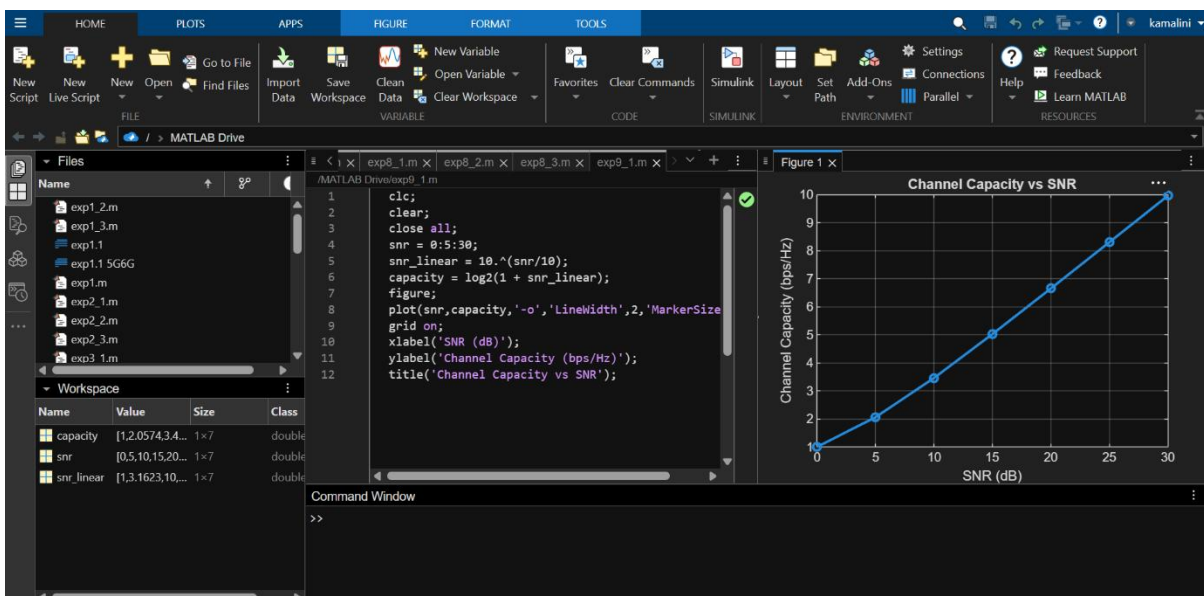
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 2:

Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB.

Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems

CODE:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

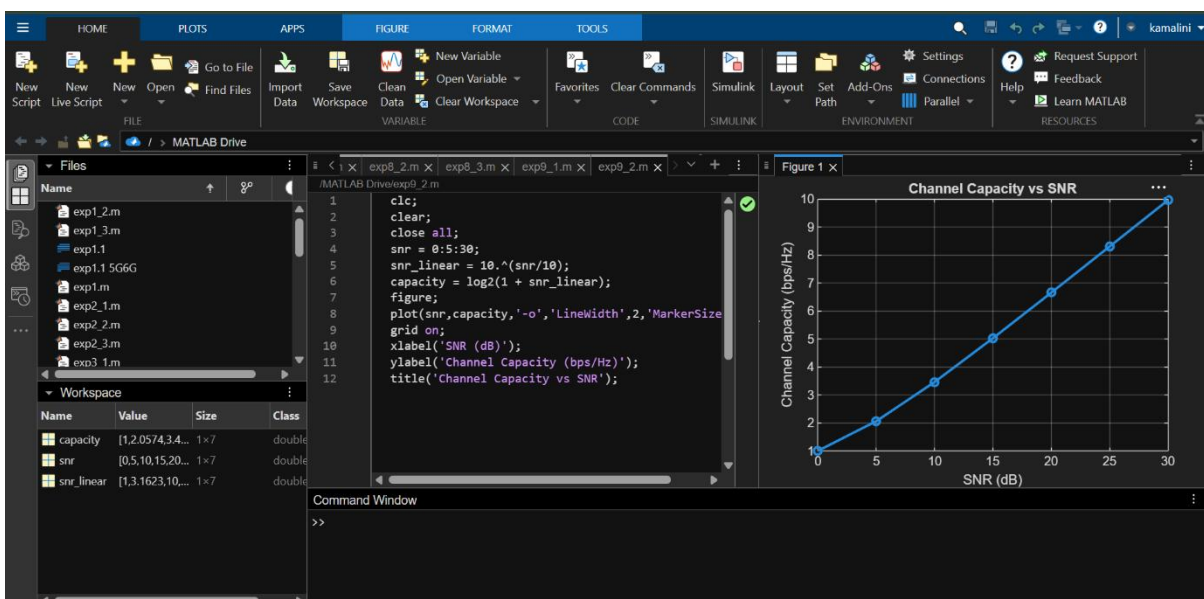
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 3:

Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

CODE:

```
clc;

clear;

close all;

antennas = [4 8 16 32 64]; % Number of Antenna Elements gain =

10*log10(antennas); % Antenna Gain (dB)

figure

bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')
```

